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Review

Solar radiation on inclined surfaces: Corrections and benchmarks

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A R T I C L E I N F O

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Predicting solar radiation on inclined surfaces is a critical task for photovoltaic energy systems design, simulation and performance evaluation. Many transposition models have been proposed in the literature; and there are abundant evaluation studies. However, these models are sometimes used incorrectly. Moreover, these errors tend to propagate through the literature. This paper aims to identify and correct some errors in the literature. It also provides a benchmark on transposition model accuracies. Twenty-six models are described using a consistent nomenclature. Model performance is ranked and pairwise accuracies are evaluated with one year of solar irradiance measurements from four locations. Although no universal model is found in this study, some are recommended. The paper comes with computer code and a small portion of experimental data.

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Nomenclature

α	circumsolar half angle [rad]
$\chi_c,\ \chi_h,\ \psi_c,\ \psi_h$	some intermediate parameters in Perez model
Δ	sky's brightness
ρ	foreground's albedo
θ	incidence angle [rad]
ε	sky's clearness, as Perez et al. (1990)
ε'	sky's clearness, as Perez et al. (1987)
$a,\ c$	sky geometry parameters, in the simplified Perez model
$a',\ b',\ c',\ d'$	sky geometry parameters, in the original Perez model
$a_0,\ a_1,\ a_2$	Muneer's parameters for the background diffuse irradiation
A_I	anisotropy index, $A_I = I_h/I_{oh}$
A_I'	alternative definition for anisotropy index, $A_I' = I/I_{sc}$
b	parameter describing the radiance distribution of overcast skies
c_β	an approximation of Revfeim's integration of the geometry associated with isotropic diffuse irradiance on a sloping surface
D_c	in-plane horizontal irradiance (DHI) [W/m^2]
D_g	irradiance due to ground's reflection (DHI) [W/m^2]
D_h	diffuse horizontal irradiance (DHI) [W/m^2]
f	Klucher's modulating factor, $f = 1 - (D_h/G_h)^2$
f'	Reindl's modulating factor, $f' = \sqrt{I_h/G_h}$
f_c'	modified Olmo's multiplying factor for anisotropic reflections
$F_1,\ F_2$	coefficients expressing the degree of circumsolar and horizon anisotropy, in the simplified Perez model
$F_1',\ F_2'$	coefficients expressing the degree of circumsolar and horizon anisotropy, in the original Perez model
f_c	Olmo's multiplying factor for anisotropic reflections
G_c	in-plane global irradiance (global tilted irradiance, GTI) [W/m^2]
G_h	global horizontal irradiance (GHI) [W/m^2]
h	solar elevation [deg]
I	normal incident direct irradiance (direct normal irradiance, DNI) [W/m^2]
I_c	in-plane direct irradiance [W/m^2]
I_h	horizontal direct irradiance ($I_h = I\cos z$) [W/m^2]
I_o	extraterrestrial normal incident irradiance (I_{sc} corrected for actual Sun–Earth distance) [W/m^2]
I_{oh}	extraterrestrial GHI [W/m^2]
I_{sc}	solar constant, 1362 W/m^2
k_t	clearness index, G_h/I_{oh}
N_{pt}	Gueymard's weighting factor between a clear and an overcast sky, or total cloud opacity
r_b	factor that accounts for direction of beam radiation, $r_b = \cos\theta/\cos z$
R_d	diffuse transposition factor
R_r	transposition factor for ground reflection
R_{d0}	diffuse transposition factor for clear condition
R_{d1}	diffuse transposition factor for overcast condition
S	Steven's anisotropy index
s	tilt angle of an inclined surface [rad]
Z	fraction of D_h that is due to collimated radiation from zenith
z	zenith of the Sun [rad]

1. Introduction

Many types of solar radiation models have been developed to predict solar radiation when measurements are not available or are inappropriate. There was not any accepted typology, until Gueymard and Myers (2008) proposed to classify radiation models based on nine criteria. A transposition model (see criterion 8 in Gueymard and Myers, 2008) typically takes measured and/or modeled irradiance components on the horizontal surface as inputs. Together with several known geometrical parameters, such as solar zenith angle, surface inclination, and incidence angle, the model predicts irradiance components on an inclined surface.

Transposition models are frequently used during photovoltaic (PV) energy systems design, simulation and performance evaluation. For example, transposition models are used to optimize the tilt and azimuth of flat-plane PV arrays (e.g., Khatib et al., 2015; Khoo et al., 2014; Lahjouji and Darhmaoui, 2013; Lave and Kleissl, 2011), so that their energy output can be maximized. Other emerging applications, such as using PV systems as sensors for spatio-temporal irradiance forecasting (see discussions in Yang et al., 2015, 2014a; Lonij et al., 2013), also require transposition models, or more specifically, the inverse transposition models (Marion, 2015; Yang et al., 2014b; Yang et al., 2013; Faiman et al., 1987).

1.1. A brief literature review of comparative and validation studies on transposition models

This work will not attempt reviewing the entire history of transposition models, but a few studies are highlighted. The present work includes minimal theoretical explanations on various transposition models; rather the focus is on the model formulations using a consistent nomenclature. In this section, a brief literature review of comparative and validation studies on transposition models is presented.

The first transposition model appeared in the 1960s (Liu and Jordan, 1963, 1961; Kondratyev and Manolova, 1960). This model is isotropic, i.e., it assumes that the diffuse radiance is uniformly distributed over the sky hemisphere. However, the anisotropic nature of sky radiance has long been demonstrated by Kondratyev and Manolova (1960) and others. Since then, a large number of increasingly sophisticated (physical and empirical) models have been proposed by treating the sky diffuse component as anisotropic (e.g., Yao et al., 2015; Muneer, 1990; Gueymard, 1987; Perez et al., 1986). On the other hand, with the technological advancements in artificial intelligence, there is a trend in using tools such as artificial neural networks to predict tilted irradiance (e.g., Ramli et al., 2015; Dahmani et al., 2014; Notton et al., 2012; Mehleri et al., 2010). Interested readers are referred to the recent review by Yadav and Chandel (2014). The scope of this paper is however limited to physical and empirical models.

Many attempts have been made to validate, compare, and review transposition models. Most early works compared the performance of slope irradiance models at daily and longer time integrals (e.g., Klein, 1977; Norris, 1966). Ma and Iqbal (1983) compared three models, with both their daily and hourly formulations, using measured data over a period of one year from Woodbridge, Ontario. Two coincident review papers (Muneer and Saluja, 1985; Hay and McKay, 1985) jointly covered most of the available models at that time.

In a report of IEA Task 9 by Hay and McKay (1988), 21 tilted irradiance models were compared. The study used 24 datasets, spanning the years from 1961 to 1984, from sites in diverse climatic regimes. It contained information about the different transposition models then available. An effective validation methodology was introduced, in which sites with different experimental setups were used to prepare separate validation datasets. A popular transposition model, the Perez model, was recommended in the concluding remarks of that report, with Gueymard being the next best model. These two models, in terms of number of parameters and granularity of modeling, are more elaborate than others; their complexity is justified by their superior performance as evident in Hay and McKay (1988). Although the remaining models were shown to be suboptimal in most test cases, they occasionally outperformed on some specific datasets. For such reasons, several less sophisticated models are included in this paper. Empirical support of the model choices is provided in Appendix A.

Throughout the past decade, the validation and comparison of transposition models has intensified. From using a handful of second generation models, according to Muneer et al. (2004)'s classification of transposition models, to 10–20 models from all three generations, many researchers have validated their models of choice with data collected at geographically dispersed locations, including Austria (Orehounig et al., 2014); Belgium (Demain et al., 2013); Brazil (Escobedo et al., 2014); Egypt (Khalil and Shaffie, 2013b); Hungary (Horváth and Csoknyai, 2015); India (Pandey and Katiyar, 2014); Israel (Evseev and Kudish, 2009b); Italy (Gracia and Huld, 2013); Northern Ireland (Mondol et al., 2008); Poland (Chwieduk, 2009); Romania (Vasar et al., 2016); South Korea (Lee et al., 2013); Spain (Posadillo and López Luque, 2009) and Thailand (Wattan and Janjai, 2016). This list is far from being complete. Nevertheless, these references, together with some other references appeared in later parts of this paper, are plotted in Fig. 1 as a word cloud. The size of a word is proportional to the number of models considered in that reference.

The growing body of recent literature on this topic, or more generally, in the areas of solar radiation and solar resource assessment, motivates the emergence of global, "all-inclusive" intercomparisons to provide more definitive answers on model performance. On this point, several (associate) editors of *Solar Energy* (Gueymard et al., 2009) have thought it advisable to communicate a short editorial regarding the journal's performance and publication criteria, in which the intention of publishing only papers containing "universal" models and/or providing unique insights was announced. Comparing transposition models with datasets from a specific site is of local and limited interest. Furthermore, in situations where only a few second generation models are compared, choosing the winning model for subsequent applications, such as evaluating PV performance, introduces additional uncertainties, which could be reduced if a superior model is used.

Uncertainty quantification is of practical importance and thus has been a topic in the literature. Loutzenhiser et al. (2007) compared 7 models using data from two 25 days periods measured in Duebendorf, Switzerland. The authors also conducted detailed

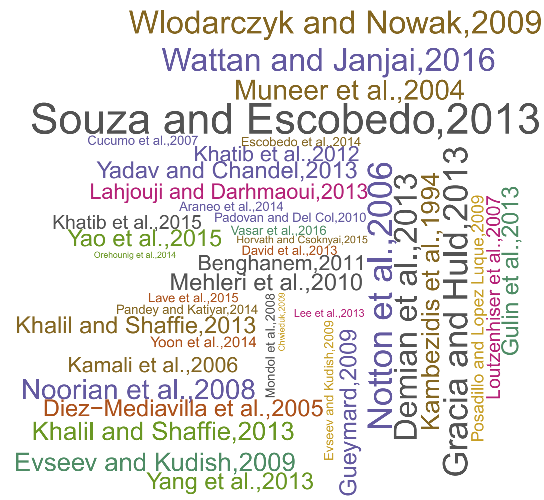

Fig. 1. Word cloud of studies that compare several transposition models. The size of a word is proportional to the number of models considered in the study, ranging from 2 to 20 models.

sensitivity analyses using Monte Carlo and fitted effects for N-way factorial analyses to assess the effect of input uncertainties on the output. The sensitivity analyses were later recommended by Gueymard and Myers (2008) in their book chapter. Gueymard (2009) conducted a study on direct and indirect uncertainties involved in transposition modeling, using 1-min data from NREL, Golden, Colorado. One of the interesting features of the work is the evaluation of 10 models using data recorded by many types of co-located instruments with different measurement uncertainties. It was found that the models which perform well using suboptimal data are not necessarily those that perform well with optimal data, due to intricate model-to-model variance in accumulation or compensation of errors (Gueymard, 2009). The uncertainties associated with foreground's albedo which enters ground reflected irradiance calculation have been investigated (e.g., Yoon et al., 2014; Demain et al., 2013; Katiyar and Panday, 2010; Gueymard, 2009).

1.2. Objectives and paper organization

In this paper, the performance of 26 transposition models are compared. Various error metrics, linear ranking, and hypothesis testing are employed to quantify the model performance. Uncertainties are not considered in this paper, but further studies on transposition modeling uncertainties are encouraged.

One contribution is that additional models compared to Hay and McKay (1988) are considered in this paper and a global benchmark on modeling accuracy is provided. However, the comprehensive literature revealed suboptimal or even incorrect usage of several transposition models. Although a poorly implemented model usually results in high errors, these suspicious numbers still sometimes slip through proofreading and peer-review. Furthermore, these errors tend to propagate through the literature. This paper therefore aims to identify some errors in the literature. The goal is not to defame the authors nor their works. Instead, it is simply to help readers to understand the origin and confusion stemming from these errors and avoid future works from repeating the mistakes. Similarly, the author welcomes comments or rebuttals on the present paper and its interpretation of model errors.

The computer programs of transposition models and quality controlled datasets were released by Hay and McKay (1988) on magnetic tape, including a user's guide and reference manual. Unfortunately, in energy journals, releasing computer code supplemental to publications is still not popular. As computer code can help avoid misinterpretations and promote the widespread uptake of the results by researchers globally, the R code used in this paper is provided as supplementary material (see Appendix F for details). While R is not the most popular programming language as of today (it was ranked 6th by *IEEE Spectrum* in 2015), it only takes a minimal effort to interface R with other popular programming languages such as Java, C, C++, Python and Matlab. The validation dataset is also provided to benchmark future implementations.

The remaining part of the paper is organized as follows. A review on 26 transposition models is first performed in Section 2. The models are validated and ranked against irradiance data on tilted planes at 4 sites with 18 different tilt and azimuth angles in Section 3. Conclusions follow at the end.

2. Transposition models

In general, a transposition model takes the form:

$$G_c = I_c + D_c + D_g, \tag{1}$$

where the global tilted irradiance (GTI) received by a solar collector (G_c) is expressed as the sum of in-plane direct irradiance (I_c), in-plane diffuse irradiance (D_c) and irradiance due to ground reflection (D_g). I_c can be calculated via:

$$I_c = I\cos\theta = I_h\frac{\cos\theta}{\cos z}, \tag{2}$$

where I is direct normal irradiance (DNI), which is the direct irradiance received by a surface normal to the sun-ray; I_h is the direct irradiance on a horizontal surface; θ is the incidence angle; and z is the solar zenith angle. While I_c is deterministically calculated, D_c and D_g are modeled by defining transposition factors:

$$D_c = D_hR_d, \tag{3}$$
$$D_g = \rho G_hR_r, \tag{4}$$

where ρ is the foreground's albedo; R_d is diffuse transposition factor; D_h is diffuse horizontal irradiance (DHI); G_h is global horizontal irradiance (GHI, $G_h = I_h + D_h$); and R_r is the transposition factor for ground reflection.

Over the years, many authors have presented transposition models. While it is common to model R_r under the isotropic assumption (Gueymard, 2009):

$$R_r = \frac{1-\cos s}{2}, \tag{5}$$

where s is the tilt angle of the inclined surface, the formulations differ by the modeling of R_d.

The transposition models are arranged in chronological order. This paper follows the naming convention used by Gueymard and Ruiz-Arias (2016), namely, first author in SMALL CAPS, optionally followed by a number if more than one version of the model is available.

2.1. Liu

Liu (Liu and Jordan, 1961):

$$R_d^{\text{LIU}} = \frac{1+\cos s}{2}. \tag{6}$$

Although this classic isotropic model was reported by Kondratyev and Manolova (1960) at an earlier time, we follow its well-accepted name.

2.2. Bugler family of models

Bugler1 (Bugler, 1977):

$$R_d^{\text{BUGLER1}} = \frac{1+\cos s}{2} + 0.05\frac{I\cos\theta}{D_h}. \tag{7}$$

Eq. (7) originates from the argument that the anisotropy in tilted diffuse irradiance should be accommodated through an additional component contributed by the circumsolar region (Bugler, 1977); Bugler assumes the addition component is 5% of DNI. However, Hay and McKay (1985) pointed out that the original formulation ignores the fact that a portion of the isotropic radiation is also contributed by the circumsolar region. Therefore, an adjusted formulation is given as:

$$R_d^{\text{BUGLER2}} = \left(1 - 0.05\frac{I_h}{D_h}\right)\frac{1+\cos s}{2} + 0.05\frac{I\cos\theta}{D_h}, \tag{8}$$

which is referred to as Bugler2 in this paper. For further information on Bugler models, the readers are referred to Appendix B.

2.3. Temps

Temps (Temps and Coulson, 1977):

$$R_d^{\text{TEMPS}} = \cos^2\left(\frac{s}{2}\right)\left[1+\sin^3\left(\frac{s}{2}\right)\right]\left(1+\cos^2\theta\sin^3 z\right), \tag{9}$$

where the $[1+\sin^3(s/2)]$ term accounts for the horizon brightening effect and the $(1+\cos^2\theta\sin^3 z)$ term accounts for the brightening in the vicinity of the Sun (the circumsolar region).

2.4. Klucher

Klucher (Klucher, 1979):

$$R_d^{\text{KLUCHER}} = \cos^2\left(\frac{s}{2}\right)\left[1+f\sin^3\left(\frac{s}{2}\right)\right]\left(1+f\cos^2\theta\sin^3 z\right), \tag{10}$$

where $f = 1-(D_h/G_h)^2$ is a modulating function to modulate Temps as the sky varies from clear to overcast.

2.5. Steven family of models

Steven1 (Steven and Unsworth, 1979):

$$R_d^{\text{STEVEN1}} = Sr_b + (1-S) \times\left\{\cos^2\left(\frac{s}{2}\right) + \frac{2b}{\pi(3+2b)}\left[\sin s - s\cos s - \pi\sin^2\left(\frac{s}{2}\right)\right]\right\}, \tag{11}$$

where $S = 0.51$ and $b = -0.87$. Steven and Unsworth (1979) also fitted S and b for different zenith angle ranges:

$$\begin{aligned} S &= 0.65, \quad b = -1.04, \quad \text{for } z = 35^\circ;\\ S &= 0.60, \quad b = -1.00, \quad \text{for } z = 45^\circ;\\ S &= 0.53, \quad b = -0.90, \quad \text{for } z = 55^\circ;\\ S &= 0.46, \quad b = -0.85, \quad \text{for } z = 65^\circ, \end{aligned} \tag{12}$$

which leads to the model Steven2, which assigns S and b in Eq. (11) according to the closest z value. An important note on the above two Steven models is given in Appendix C. Steven and Unsworth (1980) reiterated their diffuse radiation model under overcast skies:

$$R_d^{\text{STEVEN3}} = 0.143\left[\sin s - s\cos s - \pi\sin^2\left(\frac{s}{2}\right)\right] + \cos^2\left(\frac{s}{2}\right). \tag{13}$$

Eq. (13) is named Steven3. While the first two Steven models are derived for clear sky conditions, they can be used together with Steven3. In this paper, a model called Steven4 is included, which uses

Steven2 for $\varepsilon > 1.1$, Steven3 otherwise; parameter ε is sky's clearness (see Perez model below).

2.6. Hay family of models

Hay1 (Hay and Davies, 1980):

$$R_d^{\text{HAY1}} = (1 - A_I)\cos^2\left(\frac{s}{2}\right) + A_I r_b, \tag{14}$$

where $A_I = I_h/I_{oh} = I/I_o$ is the "anisotropy index" and $r_b = \cos\theta/\cos z$. The expression for A_I describes the degree of anisotropy: under a sky with complete cloud cover, $A_I = 0$, $R_d^{\text{HAY1}} = \cos^2(s/2)$ is isotropic; under a scattering free atmosphere, $A_I = 1$, $R_d^{\text{HAY1}} = r_b$ is completely directional. It should be noted that in the latter case, $D_c = r_b D_h = 0$, since $D_h = 0$ for a scattering free atmosphere. Hay (1993) gave another formulation for anisotropy index, which leads to Hay2:

$$R_d^{\text{HAY2}} = (1 - A_I')\cos^2\left(\frac{s}{2}\right) + A_I' r_b, \tag{15}$$

where $A_I' = I/I_{sc}$.

2.7. Willmott

Willmott (Willmott, 1982):

$$R_d^{\text{WILLMOTT}} = A_I' r_b + c_\beta(1 - A_I'), \tag{16}$$

where r_b is given in Hay1, A_I' is given in Hay2 and $c_\beta = 1.0115 - 0.20293s - 0.080823s^2, 0.5 \leqslant c_\beta \leqslant 1$. c_β is an approximation of Revfeim (1978) integration of the geometry associated with isotropic diffuse irradiance on a sloping surface.

2.8. Koronakis

Koronakis (Koronakis, 1986):

$$R_d^{\text{KORONAKIS}} = \frac{2 + \cos s}{3}. \tag{17}$$

2.9. Perez family of models

Perez0 (Perez et al., 1986):

$$R_d^{\text{PEREZ0}} = \frac{0.5(1+\cos s) + a'(F_1' - 1) + b'(F_2' - 1)}{1 + c'(F_1' - 1) + d'(F_2' - 1)}. \tag{18}$$

The Perez family of models (Perez et al., 1986, 1987, 1988, 1990) divides the sky hemisphere into three zones, namely, the circumsolar disc, the horizon band and the isotropic background. In Eq. (18), a' and b' are solid angles occupied by the circumsolar region and horizon band, respectively, weighted by their average incidence on a slope; c' and d' are a', b' equivalents on a horizontal surface. The coefficients F_1' and F_2' are non-dimensional multiplicative factors that relate the radiance in the two anisotropic regions to that in the main portion of the sky dome.

The transposition factor R_d^{PEREZ0} has nonlinear relationship with F_1' and F_2', which may lead to some difficulties during parameter fitting. In addition, Perez0 is more complex to use than other transposition models; it had not been validated for diverse environments (Perez et al., 1987). To that end, several modifications were made on Perez0. The right hand side of Eq. (18) was rewritten as: $0.5(1+\cos s)(1 - F_1 - F_2) + F_1(a'/c') + F_2(b'/d')$, where F_1 and F_2 are the modified multiplicative factors; they are related to F_1' and F_2' by some transformation (see Eqs. (6) and (7) in Perez et al., 1987). Furthermore, if the energy from the horizon band is assumed to be coming from an infinitesimally thin region, Eq. (18) can be written as:

$$R_d^{\text{PEREZ1}} = (1 - F_1)\frac{1+\cos s}{2} + F_1\frac{a'}{c'} + F_2 \sin s, \tag{19}$$

where

$$a' = 2(1 - \cos\alpha)\chi_c; \tag{20}$$

$$c' = 2(1 - \cos\alpha)\chi_h; \tag{21}$$

$$\chi_c = \begin{cases} \psi_h \cos\theta, & \text{if } \theta < \pi/2 - \alpha, \\ \psi_h\psi_c \sin(\psi_c\alpha), & \text{if } \theta \in [\pi/2 \pm \alpha], \\ 0, & \text{otherwise}; \end{cases} \tag{22}$$

$$\chi_h = \begin{cases} \cos z, & \text{if } z < \pi/2 - \alpha, \\ \psi_h \sin(\psi_h\alpha), & \text{otherwise}; \end{cases} \tag{23}$$

$$\psi_c = (\pi/2 - \theta + \alpha)/(2\alpha); \tag{24}$$

$$\psi_h = \begin{cases} (\pi/2 - z + \alpha)/(2\alpha), & \text{if } z > \pi/2 - \alpha; \\ 1, & \text{otherwise}, \end{cases} \tag{25}$$

$$F_1 = \max(0, F_{11}(\varepsilon') + \Delta F_{12}(\varepsilon') + zF_{13}(\varepsilon')); \tag{26}$$

$$F_2 = F_{21}(\varepsilon') + \Delta F_{22}(\varepsilon') + zF_{23}(\varepsilon'); \tag{27}$$

$$\varepsilon' = (D_h + I)/D_h; \tag{28}$$

$$\Delta = \frac{D_h}{I_o \cos z}; \tag{29}$$

and α is the circumsolar half angle, which is assumed to be 25° in Perez et al. (1987). This model is called Perez1 in this paper; its coefficients are given as the first set of coefficients in Table 1.

Perez et al. (1987, 1988) further assumed that all circumsolar energy originates from a point source. This simplification leads to Perez2:

$$R_d^{\text{PEREZ2}} = (1 - F_1)\frac{1+\cos s}{2} + F_1\frac{a}{c} + F_2 \sin s, \tag{30}$$

where

Table 1
Perez model coefficients for irradiance as a function of the sky's clearness. Note that ε and ε' are calculated differently.

	F_{11}	F_{12}	F_{13}	F_{21}	F_{22}	F_{23}
ε' (for Perez1)						
$[1, 1.056)$	−0.011	0.748	−0.080	−0.048	0.073	−0.024
$[1.056, 1.253)$	−0.038	1.115	−0.109	−0.023	0.106	−0.037
$[1.253, 1.586)$	0.166	0.909	−0.179	0.062	−0.021	−0.050
$[1.586, 2.134)$	0.419	0.646	−0.262	0.140	−0.167	−0.042
$[2.134, 3.23)$	0.710	0.025	−0.290	0.243	−0.511	−0.004
$[3.23, 5.98)$	0.857	−0.370	−0.279	0.267	−0.792	0.076
$[5.98, 10.08)$	0.743	−0.073	−0.228	0.231	−1.180	0.199
$[10.08, +\infty)$	0.421	−0.661	0.097	0.119	−2.125	0.446
ε' (for Perez2)						
$[1, 1.056)$	0.041	0.621	−0.105	−0.040	0.074	−0.031
$[1.056, 1.253)$	0.054	0.966	−0.166	−0.016	0.114	−0.045
$[1.253, 1.586)$	0.227	0.866	−0.250	0.069	−0.002	−0.062
$[1.586, 2.134)$	0.486	0.670	−0.373	0.148	−0.137	−0.056
$[2.134, 3.23)$	0.819	0.106	−0.465	0.268	−0.497	−0.029
$[3.23, 5.98)$	1.020	−0.260	−0.514	0.306	−0.804	0.046
$[5.98, 10.08)$	1.009	−0.708	−0.433	0.287	−1.286	0.166
$[10.08, +\infty)$	0.936	−1.121	−0.352	0.226	−2.449	0.383
ε (for Perez3 and Perez4)						
$[1, 1.065)$	−0.008	0.588	−0.062	−0.060	0.072	−0.022
$[1.065, 1.23)$	0.130	0.683	−0.151	−0.019	0.066	−0.029
$[1.23, 1.5)$	0.330	0.487	−0.221	0.055	−0.064	−0.026
$[1.5, 1.95)$	0.568	0.187	−0.295	0.109	−0.152	−0.014
$[1.95, 2.8)$	0.873	−0.392	−0.362	0.226	−0.462	0.001
$[2.8, 4.5)$	1.133	−1.237	−0.412	0.288	−0.823	0.056
$[4.5, 6.2)$	1.060	−1.600	−0.359	0.264	−1.127	0.131
$[6.2, +\infty)$	0.678	−0.327	−0.250	0.156	−1.377	0.251

$$a = \max(0, \cos\theta); \tag{31}$$
$$c = \max(\cos 85^\circ, \cos z). \tag{32}$$

Perez2 is parameterized by the second set of coefficients in Table 1.

The coefficients used in Perez1 and Perez2 are fitted using hourly measurements from Trappes and Carpentras, France. Although using locally-fitted coefficients can improve model performance for specific locations (e.g., Yang et al., 2014b; Gomez et al., 1992), these coefficients may not perform optimally for other locations. Therefore, Perez et al. (1990) fitted another set of coefficients using data from 10 American and 3 European sites. The model Perez3 follows the formulation of Perez2, but uses the third set of coefficients in Table 1. It should be noted that the partition of sky's clearness for Perez3 is different from that for Perez1 and Perez2. Instead of using Eq. (28), Perez3 uses:

$$\varepsilon = \frac{(D_h + I)/D_h + 1.041z^3}{1 + 1.041z^3} \tag{33}$$

to eliminate dependence between ε' and z. Perez3 is most widely accepted version of Perez model.

Utrillas and Martinez-Lozano (1994) performed a comprehensive review on various versions of Perez model. Using a dataset from València, two important conclusions were drawn: (1) the simplified circumsolar model (Perez1) performed noticeably better than the original model (Perez0), and (2) the models were sensitive to the set of coefficients used. In consideration of these, the results of Perez0 are not presented in this work. Instead, another model, Perez4, is considered, which has the formulation of Perez1 but uses the ε binning and coefficients of Perez3. Although the third set of coefficients in Table 1 could be considered as most "universal" owing to the many datasets it uses, a set of coefficients which could provide an asymptotic level of optimization is desired in the future.

2.10. Skartveit

Skartveit (Skartveit and Olseth, 1986):

$$R_d^{\text{SKARTVEIT}} = (1 - A_I - Z)\cos^2\left(\frac{s}{2}\right) + A_I r_b + Z\cos s, \tag{34}$$

where $A_I,\ r_b$ are given in Hay1 and $Z = \max(0, 0.3 - 2A_I)$.

2.11. Gueymard

Gueymard (Gueymard, 1987):

$$R_d^{\text{GUEYMARD}} = (1 - N_{pt})R_{d0} + N_{pt}R_{d1}, \tag{35}$$

where

$$R_{d0} = \exp\left[(\mathbf{XH})^\top \boldsymbol{\theta}\right] + \mathcal{F}(s)\mathcal{G}(h), \tag{36}$$
$$\boldsymbol{\theta} = \begin{pmatrix} 1 & \cos\theta & \cos^2\theta & \cos^3\theta \end{pmatrix}^\top, \tag{37}$$
$$\mathbf{H} = \begin{pmatrix} 1 & h' & h'^2 & h'^3 & h'^4 \end{pmatrix}^\top, \tag{38}$$
$$\mathbf{X} = \begin{pmatrix} -0.90 & -3.36 & 3.96 & -1.91 & 0.00 \\ 4.45 & -12.96 & 34.60 & -48.78 & 27.51 \\ -2.77 & 9.16 & -18.88 & 23.78 & -13.01 \\ 0.31 & -0.22 & -0.81 & 0.32 & 0.00 \end{pmatrix}, \tag{39}$$
$$\mathcal{F}(s) = [1 - 0.2249\sin^2 s + 0.1231\sin(2s) - 0.0342\sin(4s)]/(1 - 0.2249), \tag{40}$$
$$\mathcal{G}(h) = 0.408 - 0.323h' + 0.384h'^2 - 0.17h'^3, \tag{41}$$
$$h' = 0.01h \tag{42}$$

and h is solar elevation in degrees. While R_{d0} describes transposition factor for clear sky conditions, R_{d1} describes the transposition factor for overcast conditions:

$$R_{d1} = \cos^2\left(\frac{s}{2}\right) + \frac{2b}{\pi(3+2b)}\left[\sin s - s\cos s - \pi\sin^2\left(\frac{s}{2}\right)\right], \tag{43}$$

where $b = 1.5$ is thought appropriate (Gueymard, 1987). Under partly cloudy sky conditions, it is proposed to use a linear function to represent b, namely, $b = 0.5 + N_{pt}$, where N_{pt} is total cloud opacity, a weighting factor depending on cloud observations. If no cloud observation is available, N_{pt} can be approximated through:

$$N_{pt} = \max[\min(Y, 1), 0], \tag{44}$$
$$Y = \begin{cases} 6.6667\dfrac{D_h}{G_h} - 1.4167, & \text{if } \dfrac{D_h}{G_h} \leqslant 0.227; \\ 1.2121\dfrac{D_h}{G_h} - 0.1758, & \text{otherwise.} \end{cases} \tag{45}$$

For further information on Gueymard model, the readers are referred to Appendix D.

2.12. Muneer family of models

Muneer1 (Muneer, 1990):

$$R_d^{\text{MUNEER1}} = \begin{cases} A_I r_b + (1 - A_I)\left\{\cos^2\left(\frac{s}{2}\right) + \frac{2b}{\pi(3+2b)} \times \left[\sin s - s\cos s - \pi\sin^2\left(\frac{s}{2}\right)\right]\right\}, \\ \qquad b = -0.62, \quad \text{for } G_h > D_h,\ \theta < \frac{\pi}{2}; \\ \cos^2\left(\frac{s}{2}\right) + \frac{2b}{\pi(3+2b)} \times \left[\sin s - s\cos s - \pi\sin^2\left(\frac{s}{2}\right)\right], \\ \qquad b = 1.68, \quad \text{for } G_h = D_h,\ \theta < \frac{\pi}{2}; \\ \cos^2\left(\frac{s}{2}\right) + \frac{2b}{\pi(3+2b)} \times \left[\sin s - s\cos s - \pi\sin^2\left(\frac{s}{2}\right)\right], \\ \qquad b = 5.73, \quad \text{for } \theta \geqslant \frac{\pi}{2}, \end{cases} \tag{46}$$

where $A_I = I_h/I_{oh}$ and $r_b = \cos\theta/\cos z$.

In Muneer (1990), the $2b/\pi/(3+2b)$ term for non-overcast and unshaded conditions is empirically modeled:

$$\frac{2b}{\pi(3+2b)} = a_0 - a_1A_I - a_2A_I^2, \tag{47}$$

where $a_0,\ a_1$ and a_2 are location-dependent parameters. We name this formulation Muneer2. In the current work, the set of parameters that is fitted for the globe is considered, namely, $a_0 = 0.04,\ a_1 = 0.82$ and $a_2 = 2.026$ (Muneer et al., 2004). There are other versions of Muneer models, but their accuracies should not be dramatically different from Muneer1 and Muneer2.

2.13. Reindl

Reindl (Reindl et al., 1990):

$$R_d^{\text{REINDL}} = (1 - A_I)\cos^2\left(\frac{s}{2}\right)\left[1 + f'\sin^3\left(\frac{s}{2}\right)\right] + A_I r_b, \tag{48}$$

where $A_I,\ r_b$ are given in Hay1 and $f' = \sqrt{I_h/G_h}$. Reindl is developed based on Hay1 by including a horizon brightening correction factor similar to the one used in Temps.

2.14. Olmo family of models

Olmo1 (Olmo et al., 1999):

$$G_c^{\text{OLMO1}} = G_h\exp[-k_t(\theta^2 - z^2)]f_c, \tag{49}$$

Table 2
Data summary. CM11, CM22, CMP11 pyranometers and CH1, CHP1 pyrheliometers are manufactured by Kipp & Zonen; PSP pyranometer and NIP pyrheliometer are manufactured by Eppley; Star pyranometer is manufactured by Schenk GmbH; SPN1 pyranometer is manufactured by Delta-T; and Si-420TC-K silicon sensor is manufactured by Ingenieurbüro.

Location	Latitude	Longitude	Altitude	Albedo meas.?	DNI meas.?	Tilted sensors (tilt: s°, azimuth: β°)
Eugene	44.05	−127.07	150	Yes	Yes	(30, 180); (90, 180); (90, 0)
Golden	39.74	−105.18	1829	Yes	Yes	(40, 180); (90, 0); (90, 90); (90, 180); (90, 270)
Oldenburg	53.15	8.17	40	No	Yes	(45, 180); (45, 135)
Singapore	1.30	103.77	40	No	No	(10, 64); (20, 64); (30, 64); (40, 64); (90, 0); (90, 90); (90, 180); (90, 270)
	GHI	DHI	DNI	GTI	Period	Data link
Eugene	Calc.	Star	CHP1	PSP	2015	http://solardat.uoregon.edu/SelectArchival.html
Golden	Calc.	CM22	CH1	PSP	2014	https://www.nrel.gov/midc/apps/go2url.pl?site=BMS
Oldenburg	Calc.	CM11	NIP	CM11	2014	http://doi.pangaea.de/10.1594/PANGAEA.847830
Singapore	CMP11	SPN1	Calc.	Si-420TC-K	2013	Proprietary dataset, see Yang et al. (2014b)

where k_t is clearness index (the ratio of G_h and I_{oh}) and f_c is a factor that accounts the effect of anisotropic reflections, $f_c = 1 + \rho\sin^2(\theta/2)$. OLMO1 is different from the other transposition models in terms of parameter of interest. Instead of modeling the diffuse irradiance on the inclined surface, OLMO1 models GTI, directly. To make a consistent model representation, Eq. (49) is rewritten into:

$$R_d^{\text{OLMO1}} = \frac{G_h}{D_h}\left\{\exp[-k_t(\theta^2 - z^2)]f_c - \rho\sin^2\left(\frac{s}{2}\right)\right\} - \frac{I_h r_b}{D_h}. \quad (50)$$

OLMO1 is often found to produce significant errors, even at small tilt angles (e.g., Muzathik et al., 2011; Evseev and Kudish, 2009a; Ruiz et al., 2002). An alternative model, OLMO2, is thus proposed by Evseev and Kudish (2009a), with a new multiplying factor f_c' that is partitioned based on sky conditions:

$$f_c' = \begin{cases} 1 - \rho\cos^3(\theta/2), & \text{if } 0 \leqslant k_t < 0.35, \\ 1 - \rho\sin(\theta/2), & \text{if } 0.35 \leqslant k_t \leqslant 0.65, \\ 1, & \text{otherwise}. \end{cases} \quad (51)$$

Furthermore, the revised model is applied to diffuse radiation only:

$$R_d^{\text{OLMO2}} = \exp[-k_t(\theta^2 - z^2)]f_c'. \quad (52)$$

2.15. TIAN

TIAN (Tian et al., 2001):

$$R_d^{\text{TIAN}} = 1 - \frac{s}{180}. \quad (53)$$

The tilt angle in TIAN is in degrees.

2.16. BADESCU

BADESCU (Badescu, 2002):

$$R_d^{\text{BADESCU}} = \frac{3 + \cos 2s}{4}. \quad (54)$$

3. Validation

Before validating the 26 transposition models, we digress to discuss a recent review on separation models, which predict DHI, and thus DNI, using GHI and other observable parameters (such as cloud fraction and clear sky irradiance). Owing to the importance of separation modeling, one could immediately realize the rich literature on it. However, knowing which model could provide the highest possible accuracy at any specific location requires thorough works. In the valuable review paper by Gueymard and Ruiz-Arias (2016), 140 separation models have been validated with 1-min data collected at 54 research-class stations from 7 continents; and one may assume the result at once. More specifically, ENGERER2 (Engerer, 2015) is recommended as a "quasi-universal" 1-min separation model; under appropriate conditions (e.g., low-albedo), considering only ENGERER2 will most likely suffice. Such conclusions are hard to come by and thus are desired. However, performing studies that lead to such results is hardly an individual effort, especially when a suitable database is lacking. On this point, I would welcome any contribution of data and results that can help evolve the current work. For now, the remaining part of the paper attempts to perform a preliminary evaluation of transposition model accuracies with 4 datasets.

The 4 datasets used in this work are summarized in Table 2. As some irradiance components are measured by multiple devices, the measurements with best accuracy are identified through previous studies (Gueymard, 2009; Yang et al., 2014b) and personal communication with the respective data owners. It is believed that every attempt has been made to ensure the quality of raw data by respective owners.

As there is no optimal quality control sequence for irradiance data, I follow Gueymard and Ruiz-Arias (2016) and apply the first set of filters:

(1) $z < 85^\circ$.
(2) $G_h > 0$ and $D_h > 0$ and $I > 0$.
(3) $I < 1100 + 0.03 \times \text{Altitude}$.
(4) $I < I_o$.
(5) $D_h < 0.95 I_o \cos^{1.2} z + 50$.
(6) $G_h < 1.50 I_o \cos^{1.2} z + 100$.
(7) $|100(I\cos z + D_h - G_h)/G_h| < 5\%$.
(8) $D_h/G_h < 1.05$ for $G_h > 50$ and $z < 75^\circ$.
(9) $D_h/G_h < 1.10$ for $G_h > 50$ and $z > 75^\circ$.

Filters (1)–(9) are set mainly for irradiance quality control at the horizontal surface. For tilted data, I consider the following filters:

(10) $G_c > 0$.
(11) $0.95 > \rho > 0.05$ for Eugene and Golden datasets.
(12) Slope adjustments for G_c in Eugene and Singapore datasets.

Data points that do not satisfy the above lists are rejected. After the raw datasets (1 min resolution) are averaged into 15 min intervals, they are fed through the filters. For each location, about half of the data points (night time data points) are rejected by the $z < 85^\circ$ filter. On top of that, the combined rejection rate by all other filters is $\approx 15\%$. Normalized mean bias errors (nMBE, in percentage) and normalized root mean squares errors (nRMSE, in percentage) on the filtered datasets using the 26 transposition models are tabulated in Tables 3 and 4, respectively. A total of 18 case studies (18 different orientations/locations) are presented.

A linear ranking method (Alvo and Yu, 2014) is used to rank the models based on nMBE and nRMSE, respectively. The models are first ranked based on their performance at each orientation, i.e.,

Table 3
Normalized mean bias errors (nMBE, in percentage) using 26 transposition models on various datasets. The performance of each model is ranked for each orientation. The overall ranking of a model is then computed by averaging all of its rankings.

Model	Eugene (44.05°, −127.07°)			Oldenburg (53.15°, 8.17°)		Singapore (1.30°, 103.77°)			
	(30, 180)	(90, 180)	(90, 0)	(45, 180)	(45, 135)	(10, 64)	(20, 64)	(30, 64)	(40, 64)
Liu	2.2	3.6	−11.1	5.6	3.7	0.3	1.7	0.7	1.0
Temps	−5.3	−11.1	−30.0	−8.3	−9.6	−5.0	−4.0	−5.7	−6.8
Bugler1	−0.5	1.9	−11.2	3.3	1.5	−1.5	−0.1	−1.0	−0.8
Bugler2	2.3	4.5	−5.1	5.1	3.4	0.7	2.2	1.2	1.4
Klucher	−1.1	−3.1	−21.2	−0.7	−2.3	−2.3	−1.1	−2.5	−2.9
Steven1	−4.3	−11.9	6.4	−10.5	−10.4	0.0	1.0	−0.7	−1.4
Steven2	−4.6	−12.2	7.8	−10.8	−10.5	0.0	1.0	−0.7	−1.2
Steven3	3.1	9.1	1.5	8.0	6.2	0.5	2.4	2.1	3.3
Steven4	−1.3	−3.3	5.4	−4.3	−4.2	−0.0	1.0	−0.3	−0.5
Hay1	0.3	1.4	6.3	1.1	0.2	0.2	1.7	0.8	1.3
Hay2	0.2	1.3	7.0	1.0	0.1	0.2	1.7	0.8	1.3
Willmott	1.5	1.3	7.0	3.1	2.3	1.0	3.5	3.2	4.0
Koronakis	1.5	−6.0	−29.1	3.0	1.0	0.2	1.2	−0.5	−1.4
Perez1	−2.0	−5.0	2.6	−2.9	−3.1	−0.1	1.0	−0.1	0.1
Perez2	−2.2	−5.0	1.8	−3.2	−3.4	−0.3	0.7	−0.4	−0.1
Perez3	−1.4	−2.9	3.9	−1.9	−2.3	−0.1	1.1	0.2	0.7
Perez4	−1.2	−2.8	4.0	−1.2	−1.5	0.0	1.3	0.4	0.9
Skartveit	0.6	5.3	19.9	2.1	1.2	0.2	1.9	1.3	2.1
Gueymard	0.6	0.6	−5.2	2.2	1.1	−0.0	1.4	0.5	1.0
Muneer1	0.0	0.2	10.7	0.2	−0.9	0.1	1.2	−0.2	−0.4
Muneer2	−0.4	−2.4	9.9	−0.2	−1.3	0.1	1.1	−0.4	−0.8
Reindl	0.1	−1.8	−3.1	0.4	−0.6	0.2	1.6	0.5	0.5
Olmo1	−7.4	−17.4	−44.2	−10.2	−10.4	−1.9	−0.4	−1.3	−1.1
Olmo2	−4.6	−13.7	−6.4	−7.5	−6.8	4.5	5.8	4.6	4.5
Tian	5.9	3.6	−11.1	11.4	9.7	3.0	6.6	7.0	7.9
Badescu	4.3	3.6	−11.1	11.4	9.7	0.7	3.4	4.3	6.8

Model	Singapore				Golden (39.74°, −105.18°)					Rank
	(90, 90)	(90, 180)	(90, 270)	(90, 0)	(40, 180)	(90, 0)	(90, 90)	(90, 180)	(90, 270)	
Liu	−2.6	−13.8	−5.3	−14.3	3.9	−8.5	2.5	1.6	1.2	15
Temps	−20.7	−30.0	−23.8	−29.9	−3.0	−24.2	−9.3	−9.1	−12.4	26
Bugler1	−3.7	−14.0	−6.1	−14.5	1.1	−8.5	0.6	−0.3	−0.5	12
Bugler2	−1.7	−11.7	−4.0	−12.4	3.6	−3.2	3.2	1.9	2.7	16
Klucher	−12.0	−22.3	−14.7	−22.5	0.2	−17.8	−4.3	−4.3	−6.6	19
Steven1	−11.6	−10.9	−12.3	−9.7	−2.9	5.0	−4.2	−8.8	−6.7	18
Steven2	−10.9	−10.1	−11.6	−8.6	−3.2	6.3	−3.8	−9.0	−6.3	20
Steven3	5.9	−4.0	4.0	−4.9	5.1	1.2	7.4	5.6	6.8	17
Steven4	−6.6	−7.3	−5.7	−7.6	−0.9	4.8	−0.8	−5.0	−3.4	11
Hay1	−0.8	−6.3	−1.1	−7.2	1.0	7.7	2.3	−1.3	1.6	6
Hay2	−0.8	−6.0	−1.0	−7.0	0.9	8.3	2.3	−1.4	1.6	5
Willmott	−0.8	−6.0	−1.0	−7.0	2.1	8.3	2.3	−1.4	1.6	13
Koronakis	−16.4	−28.6	−19.9	−28.7	2.7	−23.4	−6.0	−5.5	−8.7	21
Perez1	−3.1	−5.9	−2.2	−5.7	−1.2	1.8	−1.2	−5.5	−2.9	8
Perez2	−2.8	−5.4	−1.9	−5.4	−1.4	1.2	−1.3	−5.4	−2.7	10
Perez3	−0.9	−3.4	0.0	−3.8	−0.5	3.0	0.1	−3.8	−1.4	2
Perez4	−0.6	−4.1	0.0	−4.4	−0.3	2.9	0.5	−3.7	−0.8	1
Skartveit	5.0	1.3	5.5	−0.0	1.4	15.5	5.1	0.8	5.0	14
Gueymard	−2.3	−9.3	−3.4	−10.2	1.5	−4.8	−0.0	−1.9	−1.4	7
Muneer1	−1.4	−7.3	−1.3	−7.1	0.4	7.7	1.6	−2.9	−0.0	3
Muneer2	−2.0	−7.6	−1.7	−8.0	−0.3	7.1	−0.2	−5.1	−1.5	4
Reindl	−7.0	−13.9	−8.0	−14.6	0.7	−0.3	−0.9	−3.8	−2.1	9
Olmo1	−13.2	−22.8	−16.4	−20.4	−7.4	−32.6	−9.4	−10.0	−15.5	25
Olmo2	−10.2	−10.8	−9.3	−9.8	−3.0	2.4	−5.1	−9.9	−7.3	24
Tian	−2.6	−13.8	−5.3	−14.3	7.4	−8.5	2.5	1.6	1.2	23
Badescu	−2.6	−13.8	−5.3	−14.3	6.8	−8.5	2.5	1.6	1.2	22

rank the numbers in each column of Tables 3 and 4. The overall ranking is then computed by ordering the mean ranks of the models; the mean rank $\mathcal{R}_i$ of a model i is given by:

$$\mathcal{R}_i = \sum_{j=1}^{m!} n_j v_j(i)/n, \tag{55}$$

where $v_j,\ j = 1, 2, \ldots, m!$ represents all possible rankings of the m models; n_j is the observed frequency of ranking j; $n = \sum_{j=1}^{m} n_j$; and $v_j(i)$ is the rank score given to object i in ranking j. It is found, with no surprise, that the rankings based on nMBE and nRMSE are different. Nevertheless, Perez4 performs the best in both rankings.

The above-mentioned ranking method have two deficiencies: (1) it is difficult to visualize the performance of a model against all other models in all scenarios, (2) when the errors of two models are close, we are not certain if the observed difference is significant. For these reasons, Diebold–Mariano (DM) test (Diebold and Mariano, 1995) is performed to compare the prediction accuracy of the 18 case studies. I consider a squared loss function, i.e., $g(e_t^{\text{Model A}}) = (e_t^{\text{Model A}})^2$, where $e_t^{\text{Model A}}$ denotes the prediction error for Model A at time interval t. The null hypothesis of the DM test is that the mean of the loss differential:

$$d \equiv g(e_t^{\text{Model A}}) - g(e_t^{\text{Model B}}), \tag{56}$$

Table 4
Normalized root mean squares errors (nRMSE, in percentage) using 26 transposition models on various datasets. The performance of each model is ranked for each orientation. The overall ranking of a model is then computed by averaging all of its rankings.

Model	Eugene (44.05°, −127.07°)			Oldenburg (53.15°, 8.17°)		Singapore (1.30°, 103.77°)			
	(30, 180)	(90, 180)	(90, 0)	(45, 180)	(45, 135)	(10, 64)	(20, 64)	(30, 64)	(40, 64)
Liu	5.5	14.1	31.3	11.9	13.2	4.3	7.1	11.0	13.0
Temps	8.1	17.0	46.0	12.4	13.6	7.3	7.2	10.2	11.7
Bugler1	4.7	12.3	31.1	9.7	11.2	4.9	6.2	10.1	11.6
Bugler2	4.9	13.1	30.6	11.0	11.9	3.9	6.5	10.0	11.7
Klucher	4.4	11.5	39.2	6.7	8.5	5.3	5.8	9.0	10.3
Steven1	7.6	20.3	23.9	15.1	15.6	3.7	6.1	8.6	10.8
Steven2	7.5	20.0	23.0	15.1	15.4	3.6	6.0	8.4	10.8
Steven3	6.0	17.2	27.1	13.8	14.5	4.3	7.3	11.3	13.4
Steven4	5.1	13.2	20.8	10.4	10.2	3.0	4.7	6.5	8.3
Hay1	4.0	9.1	29.1	6.8	7.2	3.0	4.5	6.8	7.7
Hay2	4.1	9.0	29.5	6.7	7.1	3.0	4.5	6.7	7.6
Willmott	4.8	9.0	29.5	7.8	7.9	3.3	5.7	7.5	8.6
Koronakis	5.0	15.4	45.4	10.5	12.4	4.3	6.9	11.0	13.1
Perez1	4.3	9.2	20.5	6.6	6.2	2.7	3.4	4.9	5.2
Perez2	4.3	9.3	19.2	6.7	6.4	2.9	3.4	4.9	5.3
Perez3	3.9	8.3	19.8	6.1	5.9	2.8	3.6	5.0	5.3
Perez4	3.8	8.4	20.1	6.3	6.1	2.8	3.6	5.1	5.3
Skartveit	4.2	11.4	32.3	7.1	7.3	3.0	4.6	6.8	7.9
Gueymard	4.2	10.7	26.1	7.4	8.0	3.3	4.7	7.2	8.1
Muneer1	4.0	10.0	25.6	6.3	7.1	3.0	4.5	7.0	8.2
Muneer2	4.4	12.2	24.5	7.2	7.8	3.1	4.4	7.0	8.1
Reindl	4.0	9.1	30.7	6.5	7.1	3.0	4.5	6.7	7.7
Olmo1	10.8	26.1	53.9	16.8	17.6	5.0	7.5	10.1	12.7
Olmo2	8.1	22.4	24.7	13.4	13.6	7.6	9.6	9.9	12.3
Tian	8.6	14.1	31.3	16.9	17.1	5.6	10.3	13.7	15.8
Badescu	7.1	14.1	31.3	16.9	17.1	4.3	7.8	12.1	15.1

Model	Singapore				Golden (39.74°, −105.18°)					Rank
	(90, 90)	(90, 180)	(90, 270)	(90, 0)	(40, 180)	(90, 0)	(90, 90)	(90, 180)	(90, 270)	
Liu	25.0	24.2	24.6	23.8	7.6	24.5	14.1	9.8	14.8	19
Temps	30.1	39.0	33.1	38.5	6.9	38.7	16.4	13.2	20.0	25
Bugler1	23.2	24.3	23.3	23.8	5.8	24.4	12.2	9.0	13.1	15
Bugler2	23.1	22.3	22.9	22.2	6.9	26.7	13.1	9.0	14.3	14
Klucher	25.7	33.3	27.8	32.3	4.4	30.7	13.0	9.5	14.7	16
Steven1	24.1	22.8	25.4	18.9	7.4	19.7	15.0	14.9	19.2	18
Steven2	23.7	23.2	25.2	18.8	7.3	19.3	14.4	14.7	18.5	17
Steven3	25.8	18.7	24.1	17.6	8.5	19.2	15.5	11.6	15.6	20
Steven4	19.7	21.2	20.4	17.7	5.1	19.4	11.4	11.3	15.7	13
Hay1	16.1	18.5	17.0	18.2	5.3	31.3	11.8	8.7	14.7	7
Hay2	16.0	18.4	16.9	18.2	5.2	31.9	12.0	8.9	15.0	6
Willmott	16.0	18.4	16.9	18.2	6.0	31.9	12.0	8.9	15.0	12
Koronakis	31.0	38.0	33.3	37.7	6.8	38.0	17.0	12.0	19.9	24
Perez1	12.6	17.3	12.8	14.9	4.3	13.8	8.5	9.0	9.2	3
Perez2	12.9	16.8	12.5	14.8	4.2	13.7	8.5	8.8	9.1	4
Perez3	12.6	16.2	12.2	14.3	4.1	14.1	8.2	7.7	8.8	2
Perez4	12.6	16.7	12.5	14.6	4.0	13.8	8.2	7.6	7.9	1
Skartveit	16.6	17.9	17.1	15.8	5.4	29.4	12.7	9.8	14.6	10
Gueymard	17.1	21.1	17.3	19.4	5.1	18.4	9.8	8.5	9.7	8
Muneer1	16.0	22.4	17.1	19.1	4.8	19.2	10.4	9.6	12.6	5
Muneer2	17.2	24.2	18.3	20.6	5.2	19.0	11.7	11.9	13.9	11
Reindl	18.0	23.7	19.7	23.3	5.0	31.3	11.7	9.2	15.0	9
Olmo1	25.6	29.5	26.7	28.1	14.2	38.5	23.5	20.9	29.7	26
Olmo2	21.7	22.3	22.8	21.3	9.0	28.7	16.9	17.6	24.1	21
Tian	25.0	24.2	24.6	23.8	10.6	24.5	14.1	9.8	14.8	23
Badescu	25.0	24.2	24.6	23.8	10.1	24.5	14.1	9.8	14.8	22

is 0, i.e., $H_0 : \mathbb{E}(d) = 0$. As either Model A or Model B could be better, a two-sided alternative is used, $H_1 : \mathbb{E}(d) \neq 0$. For each pair of models, a total of 18 DM tests are performed; the results are depicted in Fig. 2. The entries denote "Model A better than Model B" scenarios, i.e.,

$$\mathbb{E}(d) \equiv \mathbb{E}\left[g\left(e_t^{\text{Model A}}\right) - g\left(e_t^{\text{Model B}}\right)\right] < 0. \tag{57}$$

For examples, the entry "1" in the first row (from top) first column denotes that Badescu (Model A) performs better than Willmott (Model B) in 1 out of 18 case studies; the entry "16" in the last row last column indicates that Willmott outperforms Badescu 16 out of 18 times; and for the remaining test case, no evidence can be found that the two models perform differently. Fig. 2 suggests that no universal model is found (a universal model in this case would be a model that outperforms all other models in all case studies). However, it is evident that Perez family of models perform well against other models.

4. Concluding remarks

A total of 26 transposition models are arranged, discussed and validated using 18 case studies from 4 sites. Model performance comparison is conducted through two approaches, namely, linear ranking and pairwise hypothesis testing. According to the linear

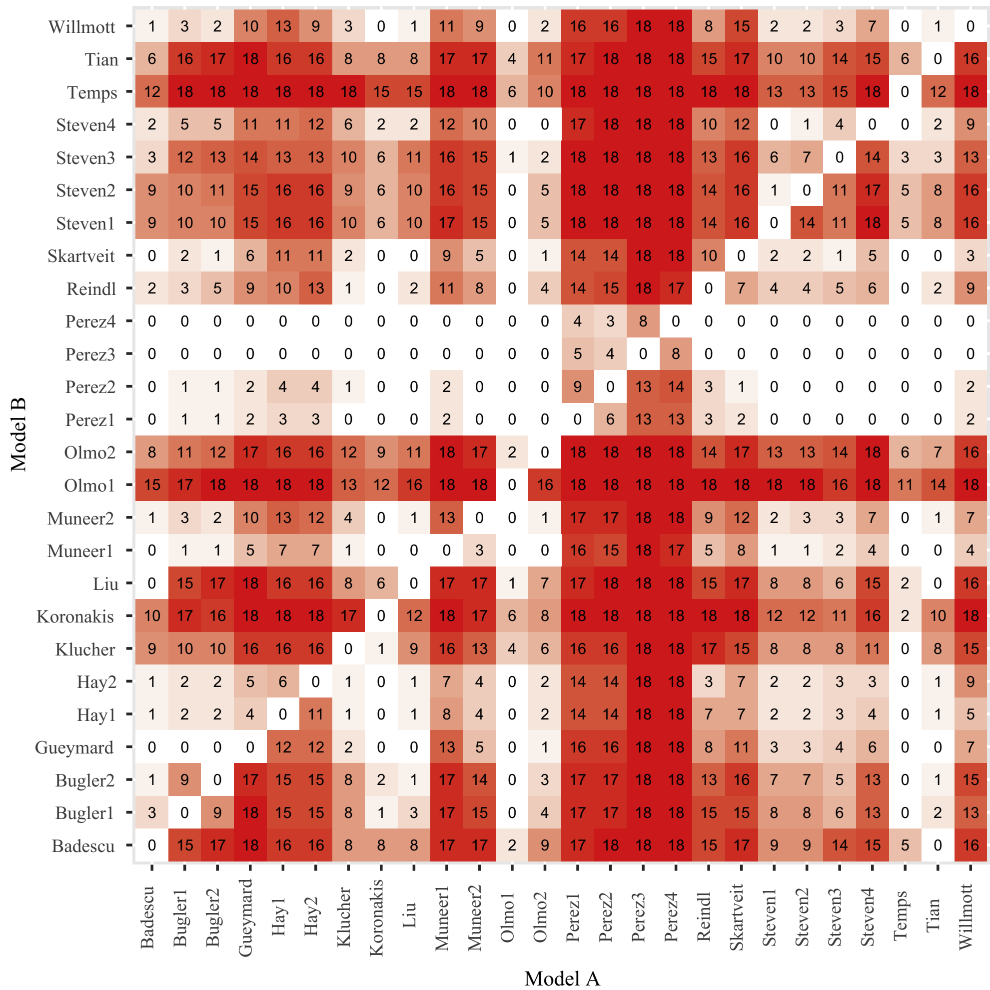

Fig. 2. Pairwise Diebold–Mariano (DM) tests for comparing the predictive accuracy of various models. For each pair of models, we count the number of instances the DM test statistics falls in the lower or upper 2.5% tail of a standard normal distribution. The entries denote "Model A better than Model B" scenarios, i.e., $\mathbb{E}(d) \equiv \mathbb{E}[g(e_t^{\text{Model A}}) - g(e_t^{\text{Model B}})] < 0$. For example, out of the 18 case studies, Badescu performs better than Bugler1 in 3 case studies (the entry is at the bottom left corner).

ranking results on nRMSE, the top four (families of) models are Perez, Muneer, Hay and Gueymard. This result agrees with the findings concluded by Hay and McKay (1988). Results of the pairwise Diebold–Mariano tests show that no universal model can be concluded. However, we are able to obtain some insights from a simple clustering analysis. Using the k-means via principal component analysis on the data matrix presented in Fig. 2, four clusters can be established: (1) Perez1, Perez2, Perez3, Perez4; (2) Gueymard, Hay1, Hay2, Muneer1, Muneer2, Reindl, Skartveit, Steven4, Willmott; (3) Bugler1, Bugler2, Liu, Klucher, Steven1, Steven2, Steven3; and (4) Badescu, Koronakis, Olmo1, Olmo2, Temps, Tian. This clustering outcome can provide a guideline on which models to choose for future studies. Based on the results of this paper, including the models out of the first two clusters is expected to provide the most accurate results.

Future work should address detailed uncertainty modeling and error analyses to further our understanding on the preferred models. For example, Tables 3 and 4 reveal that most transposition models struggle to perform in cases with vertical surfaces. As the foreground's albedo is a crucial modeling parameter in such cases, an experiment that varies the albedo could be useful to elucidate error sources.

In addition, developing more advanced transposition models with universal applicability should be attempted. The diffuse irradiance received by inclined surfaces can be calculated from angular sky radiance distribution. Most of the transposition models presented in this paper model the diffuse irradiance received by a sloped plane according to a two-part or three-part geometrical framework. However, the errors due to the approximations used in these simplified models are sometimes found to be significant (Matagne and Bachtiri, 2014; Torres et al., 2006). Therefore, calculating the tilted diffuse irradiance by integrating the radiance distribution is attractive and has great potential in moving towards a universal model. Furthermore, as mentioned by several authors (e.g., Perez et al., 1990; Gueymard, 1987), a good amount of high quality data from geographically and climatically dispersed sources is also essential for developing a universal model.

Before a universal model can be established and accepted by the community, location-specific validation studies will continue to be carried out. Although, academically, these studies are of local and limited interest, they are important to their respective subsequent solar engineering applications. As suboptimal or incorrect model implementations may lead to high errors and thus false conclusions, one should be skeptical about any unusual error terms observed during a validation study. To facilitate future studies, the R code and a small test dataset for benchmarking purposes is provided in the supplementary materials.

Conflict of interest

The author declared that there is no conflict of interest.

Acknowledgments

I would like to thank Chris Gueymard, Jan Kleissl, Elke Lorenz, André Nobre, Dave Renné and Frank Vignola for providing valuable insights and facilitating the author to obtain the data. My special thanks goes to Thomas Schmidt who helped arrange the data for Oldenburg (Kalisch et al., 2015).

Appendix A. Why are some earlier models considered by Hay and McKay (1988) not included in this study?

The 21 models considered by Hay and McKay (1988) are listed in Table A.5. As indicated in the last column of the table, 10 out of 21 models are included in the present study. There are two main reasons for not including the remaining 11 models: (1) the original study shortlisted four models (Gueymard, Hay and two versions of Perez) after a preliminary validation, while the other models were consistently outperformed by the four models; (2) empirical evidence (see below) shows that some models are less popular than others due to various reasons. This appendix focuses on illustrating the latter point.

Google Scholar citation is a reasonable indicator of popularity and impact of a publication. After searching the references listed in Table A.5 using Google Scholar, it is found that 9 out of 24 references, namely, Lawrence Berkeley Lab. (1982), Josefsson (Pers. Comm., 1985), Ineichen (1983), ASHRAE (1976), Van den Brink, ASHRAE (1971), Oegema (1971), Rogers et al. (1979) and Perez (Pers. Comm., 1985), are not tracked by Google Scholar. Despite that some of these 9 references can be found via a regular Google search, it is believed that their impacts are limited by problems such as non-English text, lengthy document and lack of digital version. They are thus not considered in the present study.

Table A.5
Hourly transposition models considered by Hay and McKay (1988). The model names follow the original publication. The third column of the table indicates whether a model is included in the present study.

Model	Reference	In this study?
Isotropic	Kondratyev and Manolova (1960)	Yes
	Liu and Jordan (1963)	
50/50 Comb	Hay (1979)	No
Bugler1	Bugler (1977)	Yes
Bugler2	Hay and McKay (1985)	Yes
	Hay et al. (1986)	
C & Z	Cohen and Zerpa (1982)	No
DOE2	Lawrence Berkeley Lab. (1982)	No
Gueymard	Gueymard (1983)	Yes
Hay	Hay (1979)	Yes
	Hay and Davies (1980)	
Hay2	Josefsson (Pers. Comm., 1985)	No
Skartveit	Skartveit and Olseth (1986)	Yes
Ineichen	Ineichen (1983)	No
Klucher	Klucher (1979)	Yes
Kusada	ASHRAE (1976)	No
Van den Brink	Van den Brink	No
Lokmanhekim	ASHRAE (1971)	No
Oegema	Oegema (1971)	No
Page	Page (1979)	No
	Rogers et al. (1979)	
Perez1	Perez et al. (1983)	Yes
Perez2	Perez (Pers. Comm. 1985)	Yes
Puri	Puri et al. (1980)	No
T & C	Temps and Coulson (1977)	Yes

The Google Scholar citation data of the remaining references are plotted in Fig. 3. Note that among the remaining 15 references, two references, namely, Gueymard (1983) and Perez et al. (1983), have more well-known versions (Gueymard, 1987; Perez et al., 1990, 1987). These more recent versions are used for citation plotting. Furthermore, Hay et al. (1986) is the errata to Hay and McKay (1985); it is thus not plotted. It can be concluded that most of these references receive increasing attention over the years, while Cohen and Zerpa (1982), Puri et al. (1980), Page (1979) are less cited. Based on this empirical evidence, together with the findings in Hay and McKay (1988), this study investigates only the selected models.

Appendix B. Notes on Bugler model

In Hay and McKay (1985), Eq. (8) was given incorrectly as:

$$D{\downarrow_s} = 0.5(D{\downarrow} - 0.05I/\cos z)(1 + \cos\alpha) + 0.05I\cos i,$$

where $D{\downarrow_s}$ and $D{\downarrow}$ are diffuse irradiance on inclined and horizontal surfaces, respectively; i and z are incidence and zenith angles; α is surface tilt and I is direct normal irradiance. The authors made a correction later and updated their results (Hay et al., 1986):

$$D{\downarrow_s} = 0.5(D{\downarrow} - 0.05I\cos z)(1 + \cos\alpha) + 0.05I\cos i.$$

Nevertheless, an incorrect equation was displayed in Kambezidis et al. (1994) (in addition to the above typo, the authors also mixed up DNI with beam irradiance in their equation), and subsequently the exact form of incorrect expression is repeated in many other works (e.g., Wattan and Janjai, 2016; Demain et al., 2013; Gulin et al., 2013; Gracia and Huld, 2013; Souza and Escobedo, 2013; Włodarczyk and Nowak, 2009; Notton et al., 2006).

On a separate note, the Bugler models used in this paper are the interpretations of Hay et al. (1986) and Hay and McKay (1985). In the original publication, DNI and DHI were unknown; they were computed *separately* using the approaches described in Sections 2.1 and 2.2 of Bugler (1977). Such treatment implies a departure from the conventional closure equation (see Eq. (3.5) of Bugler, 1977). Consequently, the global tilted irradiance derived through this approach will most likely be different from that of Bugler1. Though the model is still open to interpretation, on the account of our earlier model ranking results, further investigation seems unnecessary.

Appendix C. Notes on Steven model

Steven1 and Steven2 are not explicitly expressed in Steven and Unsworth (1979)'s original paper. This results in a large number of repetitive, incorrect interpretation of the models (such as Khatib et al., 2015; Khalil and Shaffie, 2013a; Khalil and Shaffie, 2013b; Lahjouji and Darhmaoui, 2013; Souza and Escobedo, 2013; Yadav and Chandel, 2013; Khatib et al., 2012; Benghanem, 2011; Noorian et al., 2008; Kamali et al., 2006). However, the method to deduce those empirical parameters is well documented. In this paper, I reproduce some results from Steven and Unsworth (1979) to demonstrate the correct use of Steven family of models.

By integrating the standard distribution (see Steven, 1977) of clear sky radiance, the relative diffuse irradiance (ratio between tilted and horizontal diffuse irradiance components, i.e., D_c/D_h) can be obtained. These numbers are given in Tables 1–4 of Steven and Unsworth (1979), for a range of zenith, tilt and azimuth angles. In what follows, both Eq. (11) and a commonly used wrong model:

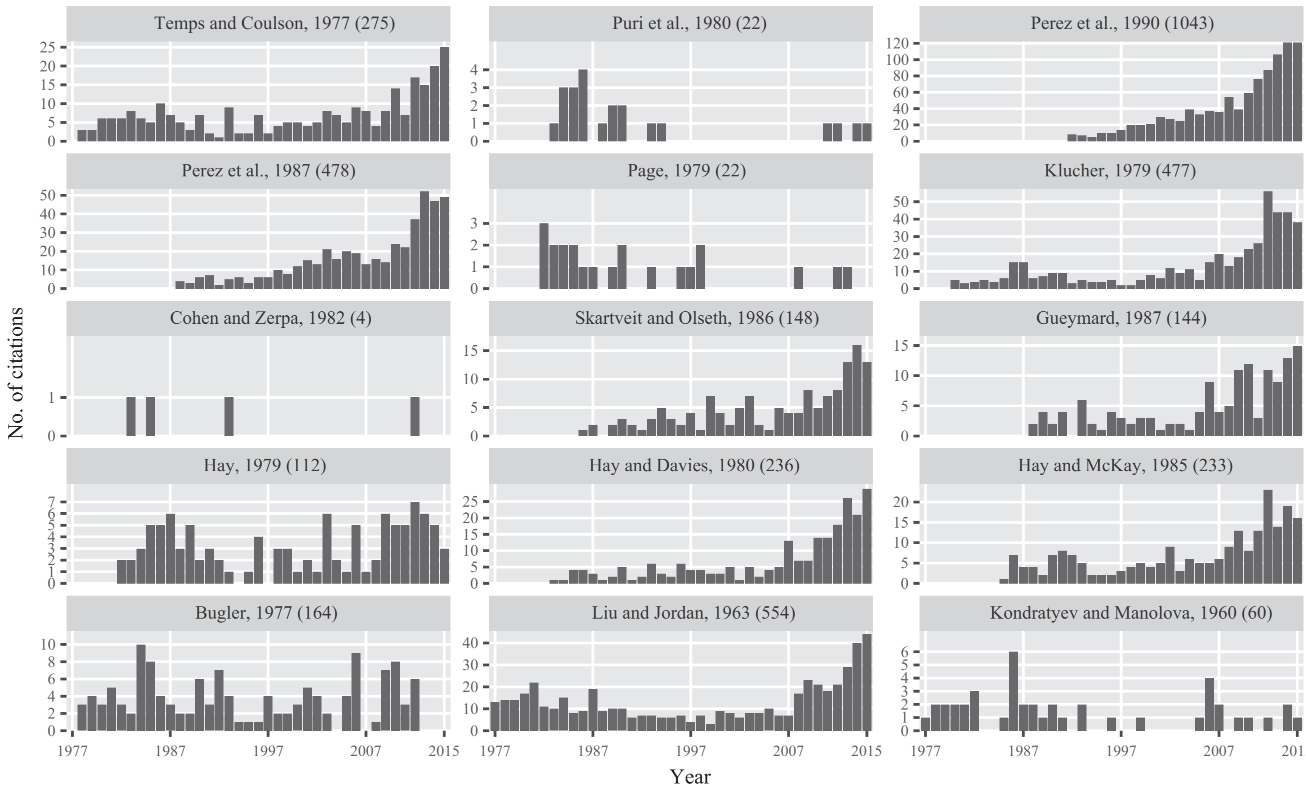

Fig. 3. Google Scholar citation data for transposition models considered by Hay and McKay (1988). Only citations after 1977 are plotted; total numbers of citations up to 2016 June are indicated in the parentheses beside the references. *Source:* Google Scholar, accessed on 2016 June 2.

Table C.6
Parameters b and S for Steven diffuse radiation models at a range of solar zenith angles (cf. Steven and Unsworth, 1979).

	Parameters	all z	35°	45°	55°	65°
Steven	b	−0.87	−1.04	−1.00	−0.90	−0.85
	S	0.51	0.63	0.60	0.53	0.46
Eq. (11)	b	−0.88	−1.09	−0.99	−0.91	−0.85
	S	0.52	0.67	0.59	0.53	0.46
Eq. (C.1)	b	7.87	3.87	5.40	6.76	7.58
	S	0.30	0.19	0.26	0.31	0.35

$$R_d = Sr_b + \cos^2\left(\frac{s}{2}\right) + \frac{2b}{\pi(3+2b)}\left[\sin s - s\cos s - \pi\sin^2\left(\frac{s}{2}\right)\right] \tag{C.1}$$

are fitted to the irradiance values in Tables 1–4 of Steven and Unsworth (1979); the fitted parameters are shown in Table C.6. It can be concluded that Eq. (11) corresponds to the original models (the fitted parameters are precise; the remaining deviation likely originates from the different non-linear least squares routines used). When the above wrong Eq. (C.1) is used, the predicted G_c is far from its actual value. The resultant high errors thus often lead to false conclusions (e.g., Noorian et al., 2008; Kamali et al., 2006).

Appendix D. Notes on Gueymard model

The original paper by Gueymard (1987) has two typographical errors. The subsequently published erratum (Gueymard, 1988) corrected the numerical values of those two coefficients. Ignoring the erratum will likely result in large prediction errors (such as Diez-Mediavilla et al., 2005). While some of these erroneous implementations have been pointed out and corrected (Diez-Mediavilla et al., 2006), others remain (e.g., Demain et al., 2013).

Appendix E. Other errors found in the references

Table E.7 shows some errors found in some references. The symbols used in the table follow their corresponding original documents; they should not be mixed with symbols defined in the nomenclature. Some of these errors are typographical and thus do not affect the results; some errors lead to wrong implementations, which can be deduced from the unusual error metrics reported. These errors may not be exhaustive.

Appendix F. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.solener.2016.06.062.

The transposition models used in this work are implemented in R (R Core Team, 2015). The code is released in the form of an R package. After unzipping the supplementary material, users should find three items, namely, a pdf document (`SolMod-manual.pdf`) which contains the package documentation, an R script (`test.R`) to generate nRMSE based on the sample data, as well as a folder (`SolMod`) which contains all the necessary files to build and install the package. Further information on building and installing R packages can be found at http://kbroman.org/pkg_primer/pages/build.html.

In addition to the implementation of the 26 transposition models, the package also performs solar positioning. The solar positioning function (`solpos`) is a wrapper over some functions in the '`insol`' package (Corripio, 2014). The example dataset

Table E.7
Some errors found in the references. Errors pointed out earlier are not repeated here.

Author	Error	Remark
Khatib et al. (2015)	Eq. (9)	Should read: $R_D = (2 + \cos TLT)/3$
	Eqs. (12) and (14)	Hay's anisotropy index should be B/I_{oh} instead of B/G
Yoon et al. (2014)	Eq. (3)	Should read $I = I_{\mathrm{DN}} \cos i + I_{\mathrm{dH}}\{(1-K)\mathrm{SVF} + K \cos i/\cos\theta_z\} + I_{\mathrm{GH}}\rho\mathrm{GVF}$
	Eq. (4)	Should read: $I = I_{\mathrm{DN}} \cos i + I_{\mathrm{dH}}\{(1-K)\mathrm{SVF} \times \mathrm{SQRT}[(I_{\mathrm{GH}} - I_{\mathrm{dH}})/I_{\mathrm{GH}}] \sin^3(\beta/2) + K \cos i/\cos\theta_z\} + I_{\mathrm{GH}}\rho\mathrm{GVF}$
	Eq. (7)	Should read: $I = I_{\mathrm{DN}} \cos i + I_{\mathrm{dH}}\{T(1-K) + K \cos i/\cos\theta_z\} + I_{\mathrm{GH}}\rho\mathrm{GVF}$
	Eq. (8)	The term 2SVF should not appear
David et al. (2013)	Eq. (6)	Should read: $\varepsilon_i = \left[(I_D + I_{B,n})/(I_D) + 5.535 \times 10^{-6}Z^3\right]/(1 + 5.535 \times 10^{-6}Z^3)$
Gracia and Huld (2013)	Eq. (37)	Should read: $f = \sqrt{G_b/G}$
Khalil and Shaffie (2013a)	Section 4	Inconsistent use of symbols. Several other errors found in this section, reappeared verbatim in Khalil and Shaffie (2013b), are listed next
Khalil and Shaffie (2013b)	Eq. (39)	Should read: $R_d = (3 + \cos 2\beta)/4$
	Eq. (41)	Should read: $R_d = (2 + \cos\beta)/3$
	Eq. (43)	Reindl's modulating factor is expressed wrongly
	Eq. (48)	Should read: $G_{d,T} = G_{d,g}\frac{1+\cos\beta}{2}\left(1 + F_1 \sin^3\frac{\beta}{2}\right)(1 + F_1 \cos^2\theta \sin^3\theta_z)$. Furthermore, Klucher's modulating factor should be: $F_1 = 1 - (G_{d,g}/G_g)^2$
Lahjouji and Darhmaoui (2013)	Eq. (14)	Should read: $R_d = (2 + \cos\beta)/3$
	Eq. (20)	Should read: $H_T = (H_g - H_d)R_b + H_g\rho\frac{1-\cos\beta}{2} + H_dR_d$
Lee et al. (2013)	Eq. (28)	Should read: $I_{D,\beta,\alpha} = I_D[f_{(\beta)}(1-K_b) + K_b \cos\mu/\sin\gamma_s]$. Furthermore, Eq. (28) should only apply to sunlit surface under non-overcast skies
	Eq. (30)	Should read: $f_{(\beta)} = \cos^2(\beta/2) + 2c/\pi/(3+2c) \times [\sin\beta - (\beta\pi/180)\cos\beta - \pi\sin^2(\beta/2)]$
Souza and Escobedo (2013)	Eq. (16)	Should read: $H^h_{R\beta} = H^h_{DH}[(2 + \cos\beta)/3]$
	Eq. (23)	Appears to be a wrong model that leads to skeptical errors terms
	Eqs. (27c) and (27d)	Should read: $Y = 6.6667K^h_{DH} - 1.1467$ if $K^h_{DH} \le 0.227$ and $Y = 1.2121K^h_{DH} - 0.1758$ otherwise, respectively
Yang et al. (2013)	Eqs. (A.5), (A.12) and (A.13)	I_{Dif} in those equations should read $I_{\mathrm{Glo}} - I_{\mathrm{Dif}}$
	Eq. (A.6)	F' in the equation should be $1 - (I_{\mathrm{Dif}}/I_{\mathrm{Glo}})^2$
	Eq. (A.7)	Should read: $\varepsilon = \left[\frac{I_{\mathrm{Dif}}+I_{\mathrm{Dir}}}{I_{\mathrm{Dif}}} + \kappa\left(\frac{Z\times\pi}{180}\right)^3\right] \div \left[1 + \kappa\left(\frac{Z\times\pi}{180}\right)^3\right]$
Khatib et al. (2012)	Eq. (10)	Should read $R_D = \frac{2+\cos TLT}{3}$
	Eqs. (13), (15) and (16)	Hay's anisotropy index in those equations should read $(E_T - E_D)/E_{extra}$
Evseev and Kudish (2009b)	Eq. (7)	Should read: $\Psi_{\mathrm{M-I}}(I_g) = k_tr_b + (1-k_t)\cos^2(\beta/2)$
	Eq. (12)	Eq. (12b) is used for a sunlit surface under a non-overcast sky. For a sunlit surface under a overcast sky and shaded surface, the b values should be 1.68 and 5.73 respectively
Włodarczyk and Nowak (2009)	Eq. (13)	The first term should read: $(1 + \cos\beta)/2$
Noorian et al. (2008)	Table 1	Hay model should read: $\frac{B_h}{G_{oh}}r_b + \left(1 - \frac{B_h}{G_{oh}}\right)\frac{1+\cos s}{2}$
	Table 1	The last term in Temps and Coulson model should read $[1 + \cos^2(\theta)\sin^3(Z)]$
Diez-Mediavilla et al. (2005)	Eq. (12)	Should read: $F_{21} + \Delta F_{22} + \theta_zF_{23}$

provided by the package contains all necessary solar positioning parameters. To use the package, the users could start with running `test.R` for a demonstration. The output from `test.R` should look like:

```
> rmse
            (40,     (90,    (90,     (90,     (90,
            180)     0)      90)      180)     270)
Liu         5.7      25.2    16.5     13.1     19.7
Temps       6.6      40.2    18.6     27.2     27.1
Bugler1     4.7      25.1    14.0     13.4     18.0
Bugler2     5.7      24.8    14.7     12.1     18.7
Klucher     5.2      33.1    15.4     21.0     20.8
Steven1     5.0      17.9    12.8     15.1     22.4
Steven2     5.3      17.2    12.5     15.4     22.0
Steven3     6.7      17.8    17.7     9.2      19.6
Steven4     4.4      17.1    10.2     14.3     18.7
Hay1        4.8      25.2    10.3     13.0     16.4
Hay2        4.8      25.5    10.4     13.1     16.6
Willmott    5.9      25.5    10.4     13.1     16.6
Koronakis   5.3      39.5    19.8     26.2     27.1
Perez1      4.2      12.9    5.9      11.9     10.1
Perez2      4.3      13.4    6.3      11.6     10.7
Perez3      4.1      13.1    6.4      10.8     10.3
Perez4      4.1      12.6    6.4      11.4     9.1
Skartveit   4.9      21.6    11.4     12.5     15.0
Gueymard    4.6      18.5    9.9      12.6     12.0
Muneer1     4.3      15.2    8.8      16.8     12.7
Muneer2     4.3      14.6    9.5      18.9     13.7
Reindl      4.5      27.4    10.7     15.7     17.5
Olmo1       8.8      32.5    21.9     27.9     29.7
Olmo2       4.5      25.9    14.6     16.2     25.4
Tian        9.7      25.2    16.5     13.1     19.7
Badescu     8.9      25.2    16.5     13.1     19.7
```

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